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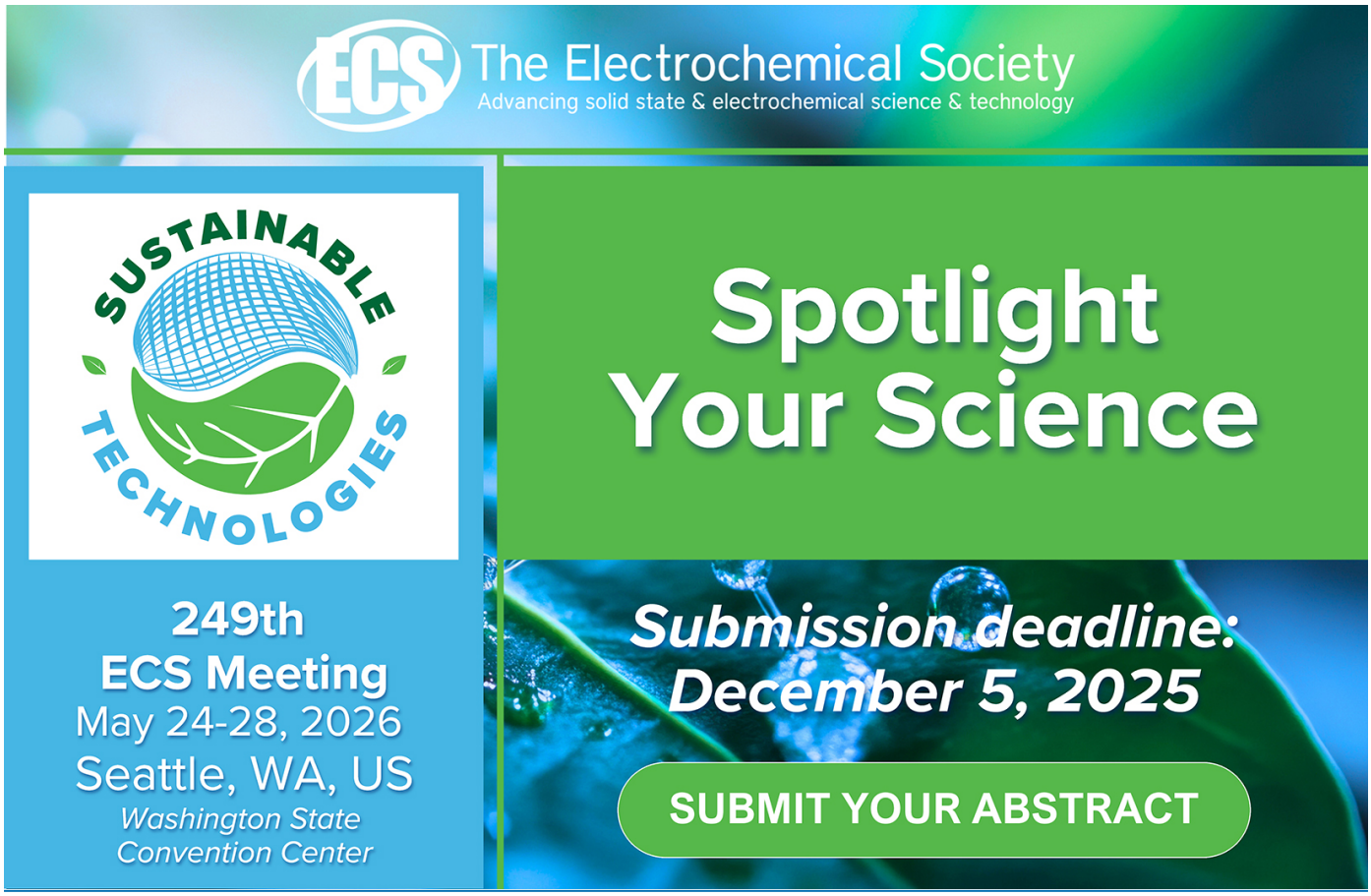

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Comparison of pharmacokinetics and biodistribution of laser-synthesized plasmonic Au and TiN nanoparticles

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Abstract. Plasmonic nanostructures offer wide range of diagnostic and therapeutic functionalities for biomedical applications. Gold nanoparticles (Au NPs) present one of the most explored nanomaterial in this field, while titanium nitride nanoparticles (TiN NPs) is a new promising nanomaterial with superior plasmonic properties for biomedicine. However conventional chemical techniques for the synthesis of these nanomaterials cannot always match stringent requirements for toxicity levels and surface conditioning. Laser-synthesized Au and TiN NPs offer exceptional purity (no contamination by by-products or ligands) and unusual surface chemistry. Therefore, these NPs present a viable alternative to chemically synthesized counterparts. This work presents comparative analysis of pharmacokinetics and biodistribution of laser-synthesized 20 nm Au and TiN NPs under intravenous administration in mice model. Our data show that Au NPs and bare TiN NPs are rapidly eliminated from the blood circulation and accumulate preferentially in liver and spleen, while coating of TiN NPs by hydrophilic polymer polyethylene glycol (PEG) significantly prolongates blood circulation time and improves delivery of the NPs to tumor. We finally discuss potential applications of laser synthesized Au NPs in SERS, SEIRA and electrocatalysis, while TiN nanoparticles are considered as promising agents for photothermal therapy and photoacoustic imaging.

1. Introduction

Plasmonic nanoparticles (NPs) is one of the most promising classes of nanosized inorganic materials for medical and biosensing applications [1]. These NPs exhibit a resonant absorption and scattering features in optical spectra with the corresponding cross-sections exceeding that of conventional absorbing dyes by many orders of magnitude [2]. The most prominent plasmonic-based therapeutic modality is

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photothermal therapy (PTT) [3,4], which profits from a conversion of external light into heat for local destruction of cancer cells and stimulation of anti-tumor immune response of the organism [5]. The same photoheating effect gives rise to a photoacoustic imaging (PAI) modality [6,7], which can be used to visualize the tumor area.

Gold nanoparticles (Au NPs) is one of the most explored plasmonic nanomaterial for biomedical applications due to chemical stability and good biocompatibility of gold. Au NPs has already been used in several diagnostic and therapeutic modalities, including PTT [8], confocal reflectance microscopy [9], photoacoustic tomography [10] and optical coherence tomography [11].

Titanium nitride nanoparticles (TiN NPs) is an alternative plasmonic nanomaterial for biomedical applications [12–14]. One of the most prominent advantages of this material is its plasmonic feature located right in the biological tissue transparency window around 650–800 nm. The feasibility of using TiN NPs as contrast agents in PAI and as sensitizers of PTT for cancer curing has already been confirmed [15].

Both of these nanomaterials are very promising candidates for plasmonic-based *in vivo* applications in nanomedicine. However, this field is very demanding to properties of the nanomaterials. The requirements include uniform size and shape, high biocompatibility, low visibility to the immune system, biodegradability or clearance from an organism [1]. Such requirements are hardly satisfied by conventional synthesis pathways. Indeed, methods of colloidal chemistry e.g., reverse micelles or the reduction of a metal precursor in the presence of capping agents [16,17] typically cause surface contamination by residual by-products and ligands, which can cause toxicity issues [18]. Moreover, Au NPs with non-spherical geometry (nanorods, nanocages, nanostars, etc.) which have biologically relevant plasmonic feature typically requires special surfactants and protecting ligands that can lead to severe toxicity effects. As an example, synthesis of Au nanorods require the use of cetyl trimethylammonium bromide (CTAB), which causes high toxicity of Au nanorods colloids [19]. On the other hand, alternative dry methods, which can be used for fabrication of TiN NPs (e.g., a direct nitridation of TiO_2 powders [20] or plasma assisted processing [21]) typically result in widely size-dispersed nanostructures of irregular shape in partially aggregated state [15,21], which complicates their stabilization in colloidal solutions, delivery *in vivo* and clearance from the body. Thus, the problem of efficient synthesis of pure uncontaminated colloids of plasmonic NPs is still actual.

We recently reported the synthesis of different types of extremely clean (free of residual contamination) colloidal NPs including plasmonic Au and TiN NPs by facile physical methods of femtosecond (fs) pulsed laser ablation in liquids (PLAL) [14,22–26]. We demonstrated that this method enables control of mean size of the synthesized NPs under relatively narrow size dispersion. Laser-synthesized NPs typically demonstrate excellent colloidal stability in absence of any stabilizing agents due to a strong negative charge of NPs leading to the electrostatic repulsion effect [27]. Moreover, our approach allows one to synthesize NPs with complex composition and geometry [28,29]. Both laser synthesized Au and TiN NPs types were thoroughly evaluated *in vivo* and *in vitro* [30-32]. The results demonstrate their good biocompatibility and favorable pharmacokinetics, which makes them very promising for the aforementioned biomedical applications.

Here, we present comparative analysis of pharmacokinetics and biodistribution of small (20-30 nm) plasmonic Au and TiN NPs in mouse model. We demonstrate that both types of NPs have similar blood circulation time, while coverage of TiN NPs by hydrophilic biopolymer polyethylene glycol (PEG) significantly prolongs that time and increase accumulation of PEG coated NPs in tumor. Both NPs preferentially accumulate in liver and spleen, but distribution among the other organs is different for Au and TiN NPs. The results confirm that covering of NPs by a hydrophilic polymers after the laser synthesis leads to prolongation of their blood circulation time and increase of their tumor accumulation.

2. Experimental methods

The NPs were synthesized by fs PLAL at ambient conditions, as described in details in our previous papers [14,30,32-34]. Au NPs we prepared using a Yb:KGW laser (1025 nm, 480 fs, 100 μJ per pulse, 10 kHz repetition rate). The laser beam was focused by 75 mm lens onto the surface of Au target

(99.99%, GoodFellow, France) placed on the bottom of a glass vessel filled with 1 cm layer of deionized water containing 1 mg/mL of dextran (MW ~40,000, Sigma-Aldrich), which was used to achieve better size control of the Au NPs and to cover them with hydrophilic polymeric coating. The ablation target was moved horizontally at 0.5 mm/s speed during the experiment. For the synthesis of TiN NPs, a TiN target was fixed vertically on the wall of a glass vessel filled with 14 mL of analytical grade acetone. Radiation from a Yb:KGW laser (1030 nm, 270 fs, 50 μJ per pulse, 10 kHz repetition rate) was focused by a 75 mm lens on the surface of the target through a side wall of the vessel. The thickness of the liquid layer from the entrance glass to the target surface was 7 mm. The ablation vessel was moved at 5 mm/s speed. A minor fraction of relatively large (~100 nm) TiN NPs were removed by centrifugation at 1000 g for 15 min. Half of the TiN NPs were covered with PEG molecules using modified Stober method. For this, 1 mg of TiN NPs was transferred to 1 mL of 96% ethanol, then 65 mL of distilled water and 20 mL of 30% ammonia hydroxide were added. Then the dispersion was ultrasonicated, and 100 mL of 1 g/L mPEG-Silane solution in ethanol was quickly dropped to it under stirring. After that, the dispersion was heated to 60 °C for 2 h followed by incubation at room temperature overnight. Finally, the PEG coated TiN NPs (TiN-PEG NPs) were separated by washing the dispersion 3 times with ethanol and one time with distilled water.

Morphology and size of the synthesized NPs were characterized by means of the high-resolution transmission electron microscopy (HR-TEM) system (JEOL JEM 3010), operating at 300 kV. Samples were prepared by dropping 5 μL of NPs solution onto a carbon-coated copper grid and subsequent drying at ambient conditions. Qualitative chemical composition was measured by an energy-dispersive spectroscopy (EDS) detector (X-act, Oxford Instruments, High Wycombe, UK) coupled with scanning electron microscope (MAIA 3, Tescan, Czech Republic) operating at 15 kV. Samples for the EDS measurements were fixed on a silicon substrate.

All experimental protocols and animal analyses for determination of pharmacokinetics and biodistribution assessment of Au NPs were conducted in accordance with the guidelines of the French Government and the Regional Committee for Ethics on Animal Experiments (authorization number 0100903). The experimental procedure was approved by the Committee for Ethics on Animal Experiments of the Institute of NeuroPhysiopathology. In the experiments with Au NPs six weeks old athymic nude female mice (Harlan, France) were used. For the pharmacokinetic study of Au NPs mice were administered intravenously with the 1 mg/kg of Au NPs. Animals were sacrificed at: 5 min, 15 min, 30 min, 45 min, 60 min and 4 h after the NPs injection and blood samples (800 μL) were collected by intra-cardiac puncture. Before the biodistribution tests human glioblastoma cells (U87-MG, 2.5×10^6) were administered subcutaneously to all mice. When the tumor reached approximately 100 mm^3, mice were intravenously administered with a single dose of 1 mg/kg Au NPs diluted in phosphate buffer saline (PBS). At the same time, control group of mice were injected with pure PBS without NPs. Animals were sacrificed at 24 h post injection. The liver, spleen, kidneys, lungs, heart, brain, tumor were removed to measure Au content. The obtained samples were mineralized followed by dilution with water and analyzed by inductively coupled mass spectrometry (ICP-MS).

All animal studies with TiN NPs were carried out in strict accordance with the provisions of the European Convention for the Protection of Vertebrate Animals Used for Experimental and Other Scientific Purposes and was approved by the Institutional Animal Care and Use Committee of Shemyakin–Ovchinnikov Institute of Bioorganic Chemistry (Moscow, Russia). Female BALB/c mice were used in the experiments. Animals bearing 300 ± 100 mm^3 mammary EMT6/P tumor were used. Blood samples for pharmacokinetic analysis were taken from the opposite retroorbital sinus. For the pharmacokinetic study of TiN NPs 200 μg of TiN NPs were injected intravenously, then 20 μL of blood was taken at 1, 5, 15, 30, 60, 120 and 180 min after the NPs injection. For biodistribution tests, mice were euthanized 180 min after TiN NPs injection. Then organs were extracted and weighted. All the samples were mineralized followed by dilution with water and measured by ICP-MS.

3. Results and discussion

Plasmonic Au and TiN NPs were synthesized by fs PLAL as described in Experimental methods section. Fabrication of Au NPs was performed in aqueous solutions of dextran polymer in order to obtain the NPs with desirable size (20-30 nm) in one step and at the same time to cover them with a polymer to increase colloidal stability in biologically relevant liquids. The possibility of *in situ* coating of the NPs during synthesis in one of the unique features of PLAL. The methodology of preparation of TiN NPs was different. The TiN NPs were originally synthesized by fs PLAL in acetone to reduce their oxidation, which negatively affects plasmonic properties of TiN NPs. Then the NPs were transferred to water and half of them were further coated with hydrophilic polymer PEG. The PEG coating was done in order to improve colloidal stability of the NPs in the bloodstream and to decrease the rate of NPs elimination by the immune system. Both of these parameters affect successful delivery of NPs to solid tumors [35]. Therefore, the PEG coating was ultimately intended to increase the accumulation of the NPs in the tumor. In total we had three NPs samples for testing their pharmacokinetics and biodistribution, namely Au NPs with dextran coating, bare TiN NPs and TiN-PEG NPs.

Physicochemical characterization of the NPs is demonstrated in figure 1.

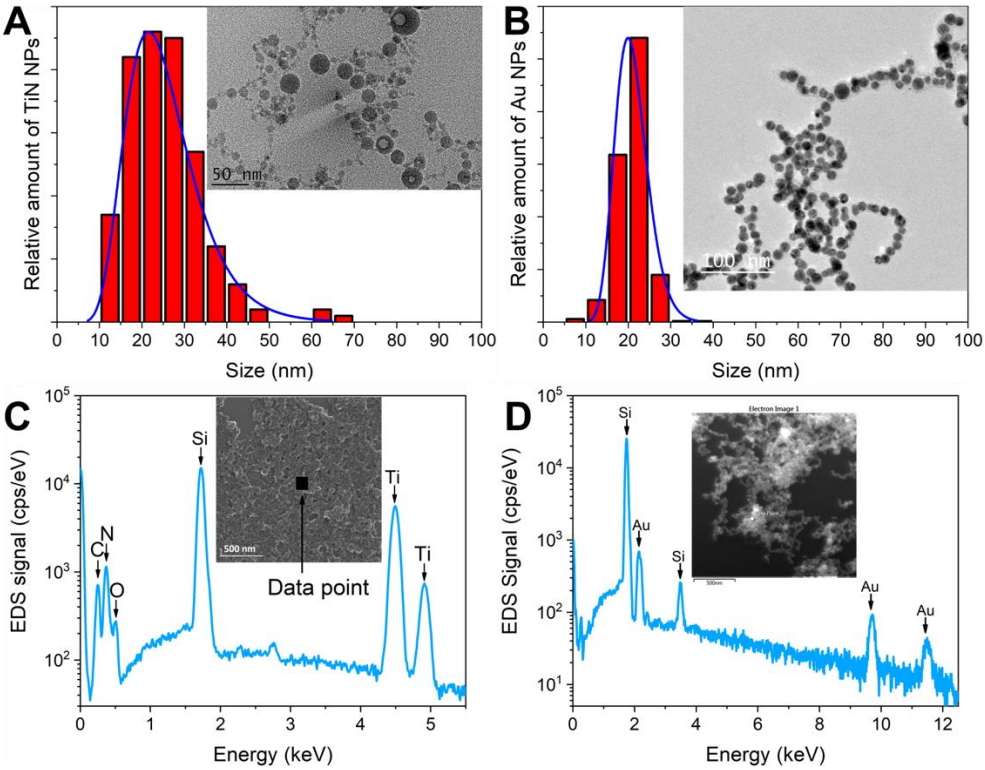

Figure 1. Physicochemical characterization of NPs. (A, B) Size distributions of TiN and Au NPs correspondingly. Insets demonstrate typical TEM images of the NPs. (C, D) Qualitative chemical composition of TiN and Au NPs correspondingly measured by EDS technique. Insets demonstrate points of spectral data collection.

Typical TEM images of TiN and Au NPs are shown in insets to figure 1A and 1B correspondingly. Morphology of both types of NPs was predominantly spherical. Statistical analysis of TEM images revealed lognormal number-weighted distributions of NPs sizes with similar mode values (19.2 nm for Au NPs and 21.5 nm for TiN NPs). This size of NPs is favorable for biomedical applications because it is small enough for effective activation of passive tumor accumulation mechanism based on EPR effect [36], but at the same time this size range is larger than glomerular filtration cut-off size and therefore

the NPs are not rapidly cleared from the blood circulation via renal filtration [37]. Note also that width of size distribution was much narrower for Au NPs than for TiN NPs (standard deviations are 3.9 and 8.9 nm correspondingly). This difference is explained by the presence of dextran in ablation liquid during synthesis of Au NPs. In our previous works we demonstrated that addition of various polymers into ablation medium can reduce mean size of NPs and narrow down their size distribution [32]. Therefore, both TiN and Au NPs had equal sizes and morphology and these parameters should equally affect biodistribution and pharmacokinetics of the NPs. EDS data demonstrated in figure 1C and 1D confirm qualitative chemical composition of the NPs. Here, in TiN NP sample the main signals in the spectrum corresponded to nitrogen (N) and titanium (Ti), while small oxygen (O) and carbon (C) signals are related to residual oxidation of the NPs and to products of acetone drying. In Au NPs only signals from Au are present, which confirm their high purity. Presence of silicon (Si) peak in spectra of both samples originated from the Si substrate.

The *in vivo* tests were performed using murine models. Tumor bearing mice were injected with NPs and pharmacokinetics along with biodistribution were studied using ICP-MS. NPs concentrations in the bloodstream were determined between 5 min and 4 h for Au NPs and between 1 min and 3 h for TiN and TiN-PEG NPs after intravenous injection of NPs. As shown in figure 2A, bare TiN NPs and Au NPs had similar pharmacokinetic profile with a short half-life time in the bloodstream and were rapidly cleared. In particular, after just 5 min after the injection we observed only 6.8% ID of TiN NPs and 1.1% ID of Au NPs in the bloodstream. However, TiN-PEG NPs had significantly improved circulation time. Five min after the NPs administration there were still more than 50% ID of TiN-PEG NPs in the bloodstream, moreover TiN-PEG NPs were detectable at substantial level even 1 h after the administration.

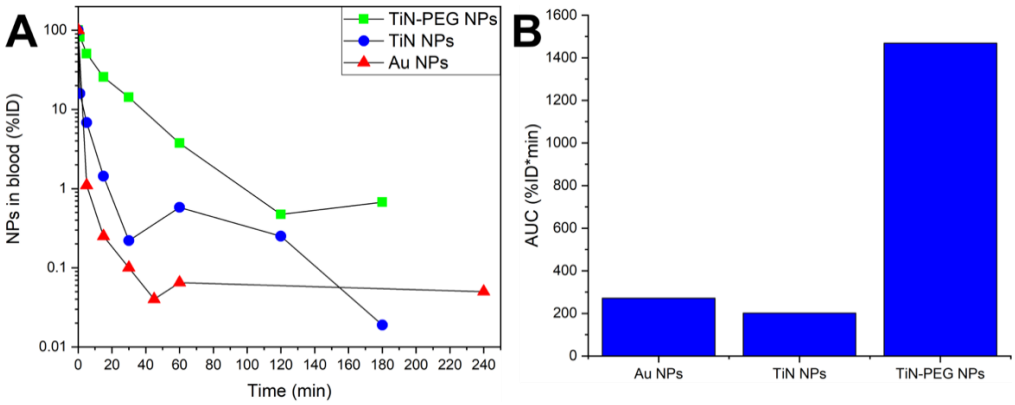

Figure 2. Pharmacokinetics data of Au, TiN and TiN-PEG NPs. (A) Blood circulation kinetics measured by ICP-MS. (B) AUC of circulation kinetics during the first 3 h after the NPs administration.

The obtained curves cannot be fitted by an exponential function due to complex pharmacokinetic behavior. Therefore, to quantitatively compare the circulation kinetics of all NPs types, we used area under the curve (AUC) parameter for the first 3 hours after the NPs injection. This parameter is commonly used in similar cases. The results are shown in figure 2B. As one can see from the data, Au and bare TiN NPs had similar AUC, while the relative parameter for TiN-PEG NPs was more than 7-fold higher. This dramatic prolongation of TiN-PEG NPs circulation can be explained by "stealth" properties of PEG polymer.

The biodistribution of all NPs was studied using data from tumor and all the major organs: liver, spleen, lungs, kidneys, heart, brain. The data were collected after all the administrated NPs were removed from the bloodstream. Results of the biodistribution analysis normalized to the maximum observed concentrations are presented in figure 3.

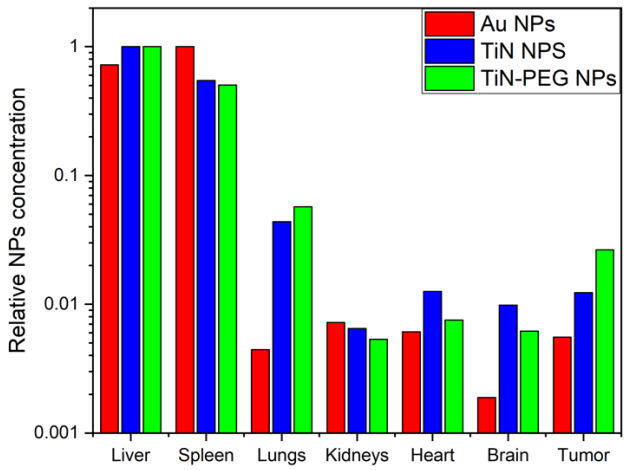

Figure 3. Biodistribution of Au, TiN and TiN-PEG NPs in organs of mice (n=3).

Direct comparison of the absolute amounts of NPs was not possible because different doses of Au and TiN-based NPs were administrated to mice. All types of NPs were predominantly accumulated in liver and spleen, however concentrations per gram of the tissue of both TiN and TiN-PEG NPs were the highest in liver, where concentration of Au NPs consisted only 72% of that in spleen. While Au NPs had the highest concentration per gram of the tissue in spleen, where TiN and TiN-PEG NPs had 55% and 50% correspondingly of their concentrations in liver. Note also that there was significantly lower relative concentration of Au NPs in lungs compared to both TiN and TiN-PEG NPs. This result can be due to presence of relatively large TiN NPs, which can get stuck in the smallest capillary in lungs. Indeed, according to figure 1A and 1B there were almost no Au NPs with sizes large than 30 nm, while substantial fraction of TiN NPs had sizes above 30 nm. Another observation that can be explained by minor size differences between NPs is slightly higher accumulation of Au NPs in kidneys. This result can be due to presence of Au NPs with sizes less than 10 nm, which are typically removed from an organism through kidney, therefore presence of smallest Au NPs leads to their higher accumulation in that organ. Finally, there was a significant difference between tumor accumulation of the NPs. TiN-PEG NPs had more than 5-fold better tumor accumulation if compared with Au NPs and more 2-fold increased tumor accumulation if compared with bare TiN NPs. This result can be due to increased blood circulation time of TiN-PEG NPs compared to other tested types of NPs. This prolongation of the blood circulation made possible their efficient delivery to tumor and accumulation via enhanced permeability and retention (EPR) effect [36]. We should note that the aforementioned differences can also be caused by the different time elapsed after the injection of NPs until the biodistribution measurements: concentrations of Au NPs were measured 24 h after the injection, while TiN and TiN-PEG NPs 3 h after the injection.

The presented data can be used for rational design of further works with plasmonic laser synthesized NPs. As an example, laser synthesized Au NPs can be of interest for a wide range of biomedical applications. These NPs can be an efficient probe for field-enhanced spectroscopies e.g., surface enhanced Raman scattering (SERS) or surface enhanced IR absorption (SEIRA). Superiority of laser synthesized Au NPs with “bare” surface for SERS has several reasons. First, exempt of ligands or contaminants eliminates parasitic background noises originated from NPs surface. Second, quenching of autofluorescence of target biomolecules from direct contact between bare surface of laser synthesized Au NPs and the biomolecules due to Foerster resonance energy transfer (FRET) effect. The advantages were demonstrated in our recent comparative tests involving laser-synthesized Au NPs and different chemical counterparts as SERS probes for the identification of beverage spoilage yeasts [38]. We found that bare laser-synthesized NPs outperform all counterparts in terms of signal-to-noise ratio. As an alternative example, we used Au-based NPs as probes for the identification of Listeria innocua and

Escherichia coli bacteria by SERS [39]. We also believe that these Au NPs can be used as promising building blocks and tags in plasmonic biosensors [40–43].

It is worth to note that the described results continue a cycle of studies devoted to the assessment of novel nanomaterials prepared by "green" laser-ablative synthesis for biomedical applications. In previous studies [44–46] we examined laser-synthesized silicon (Si) NPs which possess a series of unique imaging [47] and therapeutic [48] features. Our tests demonstrated that Si NPs are safe and biodegradable so that they can rapidly decay into orthosilicic acid and excrete from an organism with the urine within several days [46]. In contrast to widely explored Au- and Si-based NPs, which already gained substantial attention in the field of biomedical applications, TiN NPs is a new material with superior photothermal characteristics compared to other plasmonic counterparts. Although TiN NPs weakly enhance electromagnetic fields, and therefore are hardly useful in SERS, these NPs have their plasmonic peak in the window of relative tissue transparency (650-800 nm), which makes them a good candidate for PTT and PAI. We demonstrated high biocompatibility of TiN NPs and found a strong photothermal effect by using TiN NPs as sensitizers of IR PTT. Altogether, our results demonstrate that laser synthesized Au and TiN NPs are very promising for both *in vitro* and *in vivo* biomedical applications.

4. Conclusions

Biodistribution and pharmacokinetics of ultrapure fs PLAL-synthesized Au, TiN and TiN-PEG NPs with sizes in the range of 20-30 nm were measured in murine mice model. It is shown that Au and bare TiN NPs are rapidly eliminated from the bloodstream, while coating of TiN NPs with PEG increases circulation time of NPs in the bloodstream and enhance efficiency of their passive delivery to tumor target. In particular, TiN-PEG NPs had 7-fold higher AUC of their pharmacokinetic curve during first 3 h after NPs administration compared to both Au and bare TiN NPs. After systemic administration all NPs were mainly accumulated in liver and spleen, while TiN-based NPs were more concentrated in liver and Au NPs in spleen. Finally TiN-PEG NPs had 5-fold and 2-fold better accumulation in tumor than Au and TiN NPs accordingly. The presented results make laser-synthesized Au and TiN NPs very attractive for the development of novel therapeutic and imaging modalities profiting from their superior plasmonic properties.

Acknowledgements

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